

SleepSync: A Digital Intervention for Improving Sleep Behaviour in Young Adults

Sine Quinlan

22345026

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Isaiah Gitonga

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Introduction

Young Adults, or Generation Z, are in their prime developmental stage characterised by significant academic, occupational, and social demands. This involved the balance of minimum wage employment with education while simultaneously planning the trajectory of their lives. In the midst of this, the psychological burden to sustain optimal performance throughout each aspect, is foremost (Curren & Hill, 2019). This burden positions young adults as a particularly vulnerable group to the development of unhealthy sleeping behaviours, resulting in sleep disturbances that often represent three interrelated patterns. Sleep deprivation, which has been identified as an epidemic amongst young people, refers to an acute substantial loss of sleep over a short period (Medic et al., 2017). Whereas sleep restriction refers to a more chronic pattern, characterised by consistently obtaining less sleep than is biologically required across multiple nights (Banks & Dinges, 2007). While appearing distinct, these concepts frequently co-occur and contribute to circadian misalignment, a condition in which sleep-wake timing is out of sync with internal biological rhythms. In turn, this pattern contributes to a broad range of outcomes such as insomnia symptoms, excessive daytime sleepiness and mood disturbances (Baron & Reid, 2024). Moreover, such patterns contribute to impaired cognitive functioning, reduced academic performance, and vulnerability to anxiety and depression (Medic et al., 2017). Despite the widespread public health advocacy, numerous intervention attempts, and the growing body of research, irregular sleep patterns still remain highly prevalent amongst the younger generation. Improving sleep hygiene, which refers to the behavioural and environmental contributors to healthy sleep habits (Irish et al., 2015), represent a key intervention target in addressing this pattern. Therefore, SleepSync aims to improve sleep behaviours and circadian alignment in young adults through directly targeting the behavioural, cognitive and environmental contributors,

with particular focus on performance based feedback, cognitive reframing and habit formation.

Literature Review

A commonly known health guideline refers to the recommendation of approximately seven to eight hours of sleep per night that adults should obtain (Sun, 2025). However, a striking 70% of young adults fail to meet this recommendation (Hershner & Chervin, 2014).

Driven by the evident prevalence of this issue, research exploring the consequences of insufficient sleep found that more than 60% of participants reported moderate to severe levels of daytime sleepiness (Rubin et al., 2023). On a more concerning note, the inconsistent sleep cycles in young adults appear to be a major cause of increased anxiety levels, depressive symptoms, emotional instability, impaired cognitive functioning and aggression (Baglioni et al., 2011; Wang et al., 2022; Wilson et al., 2017; Van Veen et al., 2022). Furthermore, a recent review on the health impact of sleep deprivation advocates for adherence to the guideline as insufficient sleep was identified as a major risk factor for cardiovascular diseases and metabolic disorders (Shah et al., 2025).

On the other hand, high quality sleep has been associated with increased attention and concentration, alongside numerous other aspects such as memory, problem solving, emotional processing, and judgment, which are all prerequisites for daily life and learning, particularly in university students (Boyce et al., 2017; Gottlieb et al., 2018; Sio et al., 2013). In this sense, revolutionizing sleep health is vital for the health of the population.

To understand why insufficient sleep persists amongst the younger population, it is important to consider the number of factors that contribute to it. A common contributor involves conflicting societal demands and biological processes. Research found 40–60% of university students experience sleep-related problems associated with lifestyle changes such as irregular schedules, increased academic workloads, and competing social and employment

commitments (Zhou et al., 2022). Unfortunately, young adults are particularly susceptible to delayed chronotypes, a biological tendency for the body to favour later sleep and wake times (Lang et al., 2022). The circadian system is regulated by hormones such as melatonin, which promotes sleepiness (Reiter et al., 2014). In individuals with delayed chronotypes, melatonin secretion occurs later in the evening, delaying the onset of sleep (Lang et al., 2022).

However, societal demands such as early classes or work schedules demand earlier wake times than an individual's biological rhythm would prefer.

Another major contributing factor involves the current digitally driven environment. 72% of adolescents reported the use of technology in the hour before bed, which in turn impairs sleep (Gradisar et al., 2013). This late-night smart phone use is often an attempt to reclaim back personal time lost to occupational or educational demands, often a term referred to as “revenge bedtime procrastination” (Kroese et al., 2014). In this sense, sleep is often perceived as competing with personal or productive time. Interestingly, sleep disturbances linked to phone use often reflect a bidirectional relationship in which individuals use devices to fill time or regulate emotions when experiencing difficulty falling asleep, in turn delaying bedtimes sometimes over an hour (Bauducco et al., 2024).

This pattern may be understood through a concept known as temporal discounting, whereby individuals favour immediate rewards over delayed benefits (Story et al., 2014). For instance, smartphone use provides immediate gratification through stimulating content, notifications and social connectedness, which are often deemed better than the distant future benefits of sufficient sleep.

The COM-B model (Michie et al., 2011) may provide further understanding of sleep behaviours, through the core concepts of capability, opportunity, and motivation. In this context, individuals may lack psychological capability (e.g knowledge of sleep hygiene), face environmental barriers such as late-night digital media use and social demands (opportunity),

and prioritise immediate rewards over long-term benefits due to habitual processes like bedtime procrastination (motivation).

With these behavioural and environmental insights, common recommendations for improving sleep include maintaining consistent sleep–wake timing, avoiding stimulants before bedtime, reducing evening exposure to devices, and creating an appropriate environment for sleep (Irish et al., 2015). However, adherence remains consistently low, and the problem persists. Consequently, research has increasingly shifted towards the development of interventions aimed at improving sleep habits and circadian alignment within young adults. Even so, several challenges remain across the literature.

A systematic review examined the effectiveness of sleep education programs in improving general sleep hygiene practices among college students aged 18 years and older. However, the findings were inconsistent and contrasted across the studies. Sleep quality outcomes were also limited and three of the four studies reported no significant improvement (Deitrich et al., 2014). Although increasing knowledge is an important aspect of behaviour change, the findings highlight an important limitation of interventions that rely solely on an educational and informative approach.

Furthermore, interventions that require intensive behavioural change can be difficult for individuals to maintain and techniques that involve strict regulation of sleep schedules have been seen to lead to reduced adherence (Kyle et al., 2014). Additionally, Mispel et al. (2017) examined engagement with digital sleep interventions and found higher attrition rates among participants required to complete frequent questionnaires compared to those engaging with more interactive components.

The increasing popularity of wearable sleep tracking devices reflects growing public interest in using technology to monitor and improve sleep, with approximately 50% of consumers reporting willingness to purchase such devices (Baron et al., 2017). These

wearables are often used with the expectation that they can help identify and resolve sleep problems. However, Baron et al. (2017) introduced the unintended consequences of these devices, referring to the concept of “orthosomnia”. Orthosomnia is a maladaptive preoccupation with achieving “perfect” sleep based on tracker data, which can increase anxiety and reduce sleep efficiency. This highlights a key limitation of data-driven sleep interventions, as without accompanying behavioural guidance or psychological support, users may misinterpret sleep data and develop unhelpful beliefs that ultimately reinforce maladaptive sleep behaviours.

Taken together, past interventions highlight key areas for improvement, particularly the need to address productivity pressures, bedtime procrastination, and late-night digital media use as drivers of delayed sleep. In response, the present intervention, SleepSync, proposes a digital intervention designed to support healthy sleep behaviours. By targeting bedtime procrastination, technology use, and perceptions of sleep as “unproductive,” it aims to promote sustainable sleep habits among young adults and in turn, overall wellbeing.

Stakeholder Engagement

Stakeholder engagement is central to the development of SleepSync. Primary stakeholders include young adults (18-25) as the target population. Additional stakeholders include wellbeing services, health psychologists, and digital health developers, who support intervention design and implementation. Government bodies and institutional stakeholders, such as universities and employers, are also highly relevant due to their role in funding, and influences on sleep behaviour.

SleepSync values collaborative development with the target population. Guided by a co-design approach, focus groups will be conducted to gain direct insight into the experiences, needs and perspective on sleep behaviours and interventions. These focus

groups aim to identify practical challenges such as why young adults struggle to value good sleep and adhere to sleep interventions, as well as the types of features they find most engaging and beneficial within an app.

Insights drawn from existing literature and case studies examining digital sleep interventions will further identify stakeholder needs and support the successful implementation of SleepSync. Analysing what works and what doesn't allows for an informed development process and promotes user centered design by incorporating the experiences of the end users.

As outlined previously, education-based approaches and rigid protocols have shown limited effectiveness, highlighting the need for more engaging and user-centred solutions. Dennison et al. (2013) emphasises the importance of low-effort and immediate benefits in apps, with a value for features such as tracking behaviours. Furthermore, interventions delivered via mobile applications that incorporate approaches such as CBT-I, mindfulness training, and digital sleep education, identified improvements in sleep quality, sleep efficiency, and sleep hygiene, as well as reductions in insomnia severity and dysfunctional beliefs about sleep. Notably, these benefits were maintained at follow-up, indicating sustained improvements over time (Lu et al., 2025).

Taking these insights into consideration and further guided by stakeholder input, SleepSync will focus on features that align with young adults' preferences for accessible, flexible, and engaging features. In doing so, it aims to address common barriers to adherence, such as low motivation and disengagement, while supporting sustainable improvements in sleep behaviours.

Intervention and Theory of Change

A visual representation of SleepSync is provided in Appendix A.

Building on the limitations identified in existing interventions and the behavioural mechanisms underlying sleep disruption, SleepSync is designed as a theory-driven, digital intervention targeting the key drivers of poor sleep in young adults. Alongside incorporating key principles from Cognitive Behavioural Therapy for Insomnia (CBT-I) such as stimulus control, sleep hygiene, and cognitive restructuring (Walker et al., 2023), SleepSync is also grounded in the COM-B model of behaviour change (Michie et al., 2011) and self-regulation theory, which emphasises self-monitoring, feedback, and behavioural adjustment (Bandura, 1991). Taken together, the present intervention aims to target low behavioural awareness, reduced motivation driven by temporal discounting, and the lack of circadian-aligned sleep habits, while leveraging smartphone use as a platform for behaviour change (Bauducco et al., 2024). The intervention involves three core mechanisms:

1. Behavioural Awareness

We recognise that while many individuals realise that they are “not sleeping well,” they don't deem this reason enough to sacrifice the inconsistent bedtimes, prolonged screen exposure, or irregular wake times. Drawn from the principle of the self regulation theory (Bandura, 1991), SleepSync incorporates a behaviour-based tracking feature that prompts users to log key behavioural inputs, including bedtime consistency, evening screen use, and engagement in wind-down routines, which are then linked to personalised feedback and behaviour–outcome projections, highlighting future impact. Unlike modern sleep trackers that emphasise sleep metrics and outcomes that may inadvertently promote anxiety and maladaptive monitoring behaviours, SleepSync provides awareness to the user on the behaviours that influence sleep. SleepSync hopes this feature will take the focus away from abstract sleep scores, and push it towards behaviourally actionable data. This feature is further supported within the COM-B framework by increasing psychological capability by helping users understand what to change and why.

2. Cognitive reframing

A defining feature of sleepSync addresses the perception of sleep being “unproductive” or lost time (Kroese et al., 2014), a belief that is particularly relevant amongst young people. SleepSync incorporates a “Reframe” component alongside daily performance-linked feedback, designed to reposition sleep as an active contributor to next-day functioning rather than a passive state of rest. First, the user will complete a small reaction time or puzzle-based task, this is then combined with the user’s previous night’s sleep behaviours (e.g., sleep timing, consistency, wind-down behaviours). In line with evidence that digital interventions are more engaging when they are low-effort, interactive, and provide immediate benefits (Dennison et al., 2013; Mispel et al., 2017), SleepSync provides immediate, personalised feedback, linking these behaviours directly to next-day functioning, reframing good sleep as a productive, beneficial habit. For instance, if the user performed poorly on the task, and slept for 5 hours the previous night, they might see:

“Your slower response time today may reflect inconsistent sleep timing. Consider pacing your workload and prioritising recovery tonight”

On the other hand, if the user performed well, they might receive a prompt such as:

“Your consistent sleep timing is supporting strong cognitive performance today, good payoff!.”

This reframe feature is not education or sleep tips, but simply a reinterpretation of sleep as performance fuel. This overcomes the limitations identified in previous interventions that focus primarily on information and knowledge, rather than telling users that sleep is important, SleepSync enables them to experience its value directly. In COM-B terms, this directly targets reflective and automatic motivation, reinforcing sleep as behaviour worth prioritising. Furthermore, by explicitly linking sleep behaviour to next-day cognitive output,

the intervention reduces temporal discounting by converting delayed health benefits into immediate, observable outcomes, allowing users to experience the effects of their sleep more immediately and meaningfully (Story et al., 2014). This increases the perceived value of sleep, making it more competitive with immediately rewarding behaviours such as late-night digital media use.

3. Habit Formation and Circadian Alignment

Finally, the intervention focuses on building better sleep habits in a way that fits into real life. These include structured wind-down routines, reduced late-night digital engagement, and consistent sleep–wake scheduling. By supporting incremental behavioural changes rather than imposing overwhelming sleep protocols, sleep hygiene becomes more compatible with the irregular lifestyles of young adults.

This feature is supported through a circadian-informed system of timed, personalised prompts, aligned with the user’s biological rhythms. For instance, users receive reminders to start winding down, reduce screen exposure, or adjust light levels at biologically relevant times, targeting both the behavioural and physiological drivers of sleep disruption.

In order to lower the cognitive burden associated with initiating sleep-promoting behaviours (Harvey, 2002), structured, low-effort activities such as guided breathing, audio content, or progressive disengagement from screens, are offered within the app. Over time, repeated exposure to these cues supports habit formation, allowing sleep-related behaviours to become more automatic. Within the COM-B framework, this strengthens automatic motivation while also gradually modifying the user’s environment and providing the opportunity to make healthier behaviours easier to perform. This way sleep-related behaviours become increasingly automatic and less dependent on conscious motivation. Importantly, contrasting with rigid sleep restriction protocols past interventions have applied,

SleepSync supports gradual adjustment and flexibility, enabling users to shift their sleep timing incrementally in a way that is both realistic and sustainable.

Implementation and Evaluation

The implementation of SleepSync is guided by co-design principles, incorporating continuous input from key stakeholders to ensure feasibility, engagement, and real-world applicability. Insights from focus groups will inform core design decisions, including preferred feedback styles, optimal levels of interaction, and features most likely to sustain long-term engagement. In collaboration with digital developers, these insights will be transformed into a user-friendly interface. The app will adopt a simple structure, with accessible core features such as behavioural tracking, daily cognitive tasks, and performance feedback, alongside a minimalistic design using calm aesthetics to support its sleep-focused purpose (See appendix A).

Informed by past research on application implementation and evaluation (Oliviera et al., 2022), usability will be evaluated through repeated user testing to ensure the intervention is functional, engaging and practical. In parallel, consultations with sleep researchers and health psychologists will ensure alignment with the evidence-based principles applied.

A four-week pilot study will then be conducted, with participants recruited through social media, university platforms, and wellbeing services. During this period, users will engage with key features of the intervention, including behaviour tracking, cognitive tasks, circadian-aligned prompts, and performance-based feedback.

Alongside continued user testing, to further evaluate the effectiveness of SleepSync, data will be collected at baseline (Week 0) and post-intervention (Week 4). Assessment of

changes in sleep behaviour, cognitive functioning, and psychological outcomes will then be carried out through the following mechanisms:

Sleep Behaviours

The Pittsburgh Sleep Quality Index (PSQI) (Buysse et al., 1989) will be used to evaluate overall sleep quality, while app-derived data will capture sleep timing consistency and behaviours.

Cognitive Performance

Cognitive performance will be assessed through in-app tasks measuring reaction time, attention, and simple problem-solving ability. These metrics serve both as an intervention component and an outcome measure, allowing analysis of whether improvements in sleep behaviours are associated with enhanced daytime functioning.

Psychological wellbeing

Psychological wellbeing will be assessed using brief self-report measures of mood, perceived energy, and daily functioning.

Engagement and usability metrics

Engagement and usability metrics will also be collected, including frequency of app use, completion rates of daily tasks, and interaction with specific features. Post-intervention feedback surveys and optional follow-up interviews will provide insight into user experience, perceived usefulness, and barriers to engagement. Crucially, user satisfaction will be considered a critical indicator of feasibility, particularly given the known challenges of adherence in digital health interventions.

Ethical Considerations

Although SleepSync is grounded in theory and evidence based approaches, key ethical issues should be taken into consideration.

Firstly, the potential for performance-related pressure raises concern, arising from the intervention's framing of sleep as a driver of cognitive functioning and productivity. While intended to enhance motivation, users may begin to interpret sleep and performance in rigid, outcome-driven terms, leading to self-criticism or anxiety when performance is suboptimal. Additionally, daily cognitive tasks may introduce a risk similar to orthosomnia, encouraging preoccupation with achieving "optimal" scores and shifting engagement towards perfectionistic monitoring rather than reflective self-improvement. To address this, SleepSync will emphasise variability and self-compassion within feedback, framing performance as informational rather than evaluative. Messaging will avoid comparative language and instead highlight trends over time, reducing the likelihood of perfectionistic engagement.

A second consideration concerns data privacy. SleepSync collects behavioural data related to sleep patterns, daily routines, and cognitive performance, all of which may be considered personal and potentially sensitive. Therefore, the intervention will adhere to data protection regulations such as GDPR and users will be clearly informed about what data is collected, how it is used, and their rights regarding access, withdrawal, and deletion.

A social justice consideration relates to accessibility. Although sleepSync is a digital intervention delivered via smartphones, some individuals may not have equal access to such technology, reliable internet or digital skills (Wilkin et al., 2017). To address this, SleepSync will be designed to be low-data, compatible across all devices, and have a simple and intuitive interface to reduce digital skills barriers.

Finally, there is a risk related to over-reliance on the application. While SleepSync is designed to guide behaviour through prompts and structured routines, users may become dependent on these external cues to regulate their sleep behaviours. To address this, SleepSync will incorporate features that encourage users to attend to internal cues, such as

reflective prompts that ask users to assess their own tiredness levels alongside app suggestions. Additionally, the frequency of prompts may be gradually reduced over time as users establish more stable routines, supporting the transition from externally guided behaviour to autonomous habit formation.

Appendix

Appendix A

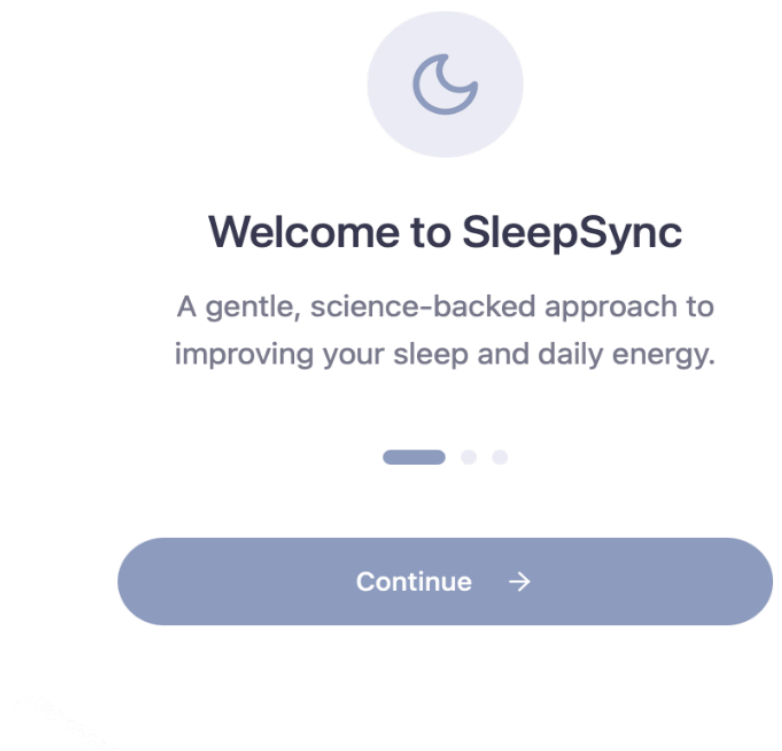


Figure A1. SleepSync onboarding screen

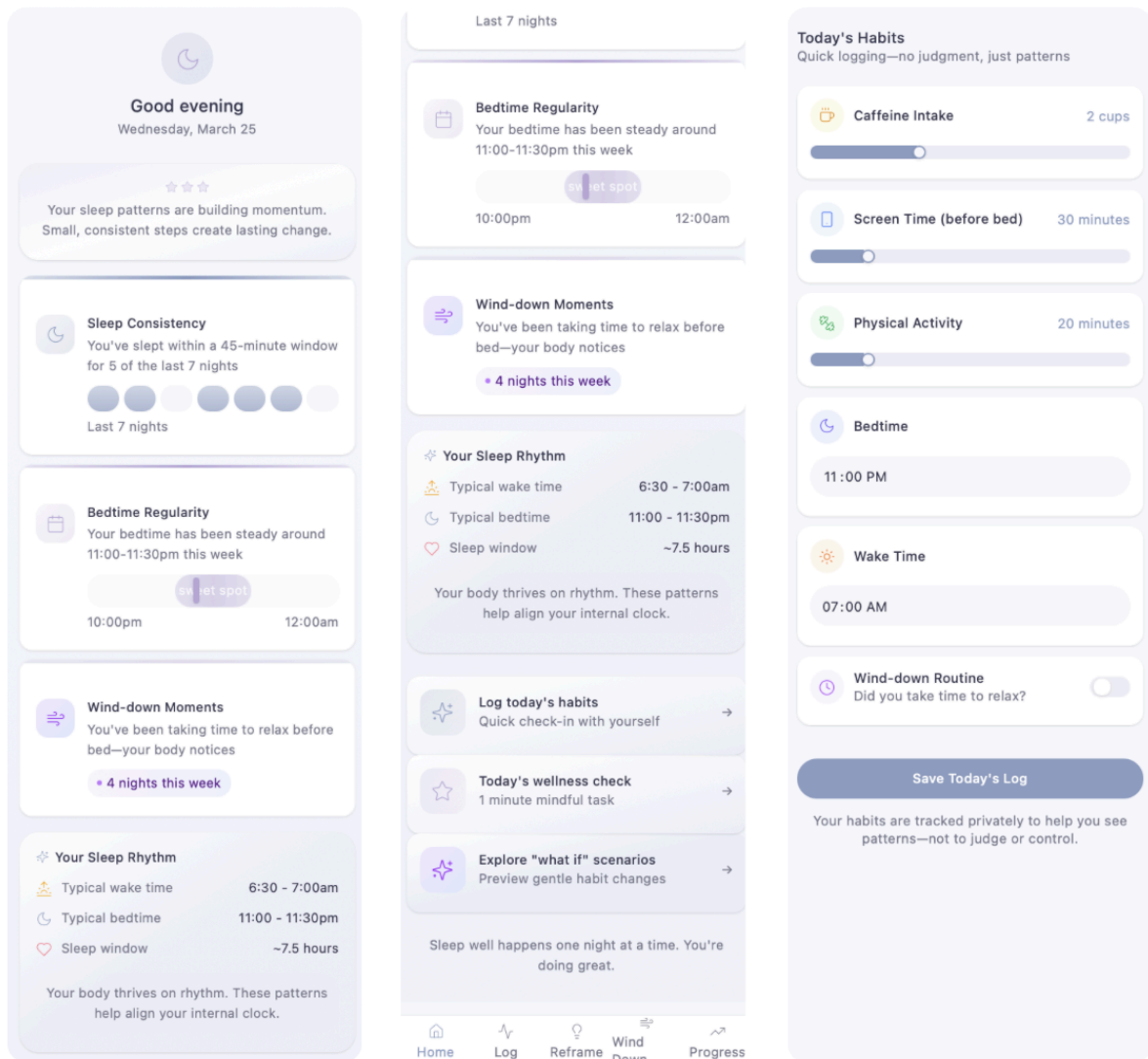


Figure A2. SleepSync home dashboard and daily behaviour logging interface

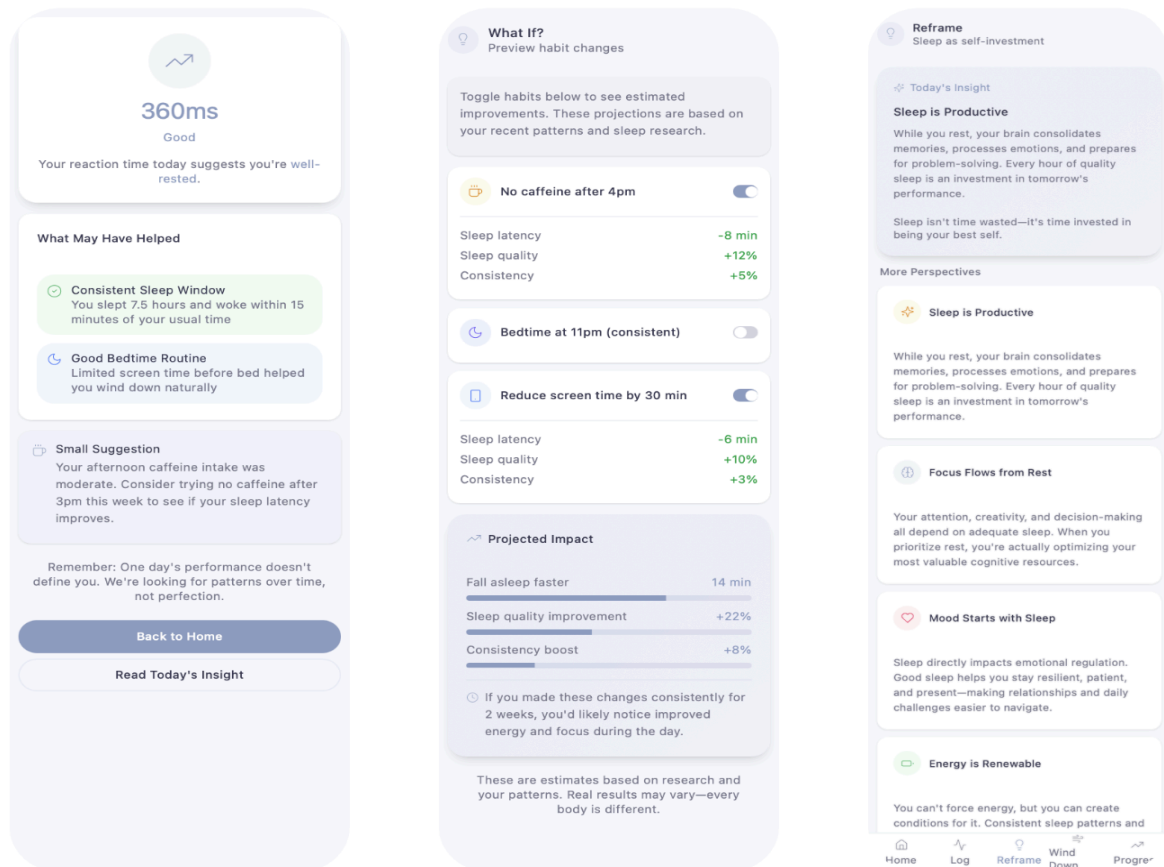


Figure A3. SleepSync cognitive performance feedback, behaviour–outcome projections, and cognitive reframing interface

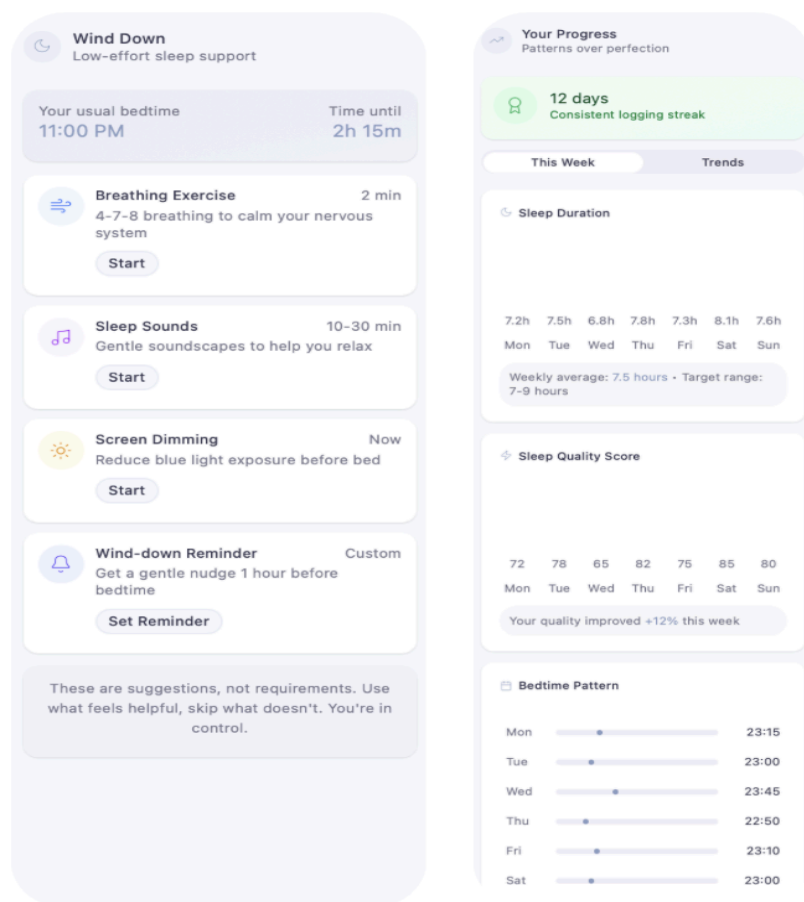


Figure A4. SleepSync wind-down support and progress tracking interface

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